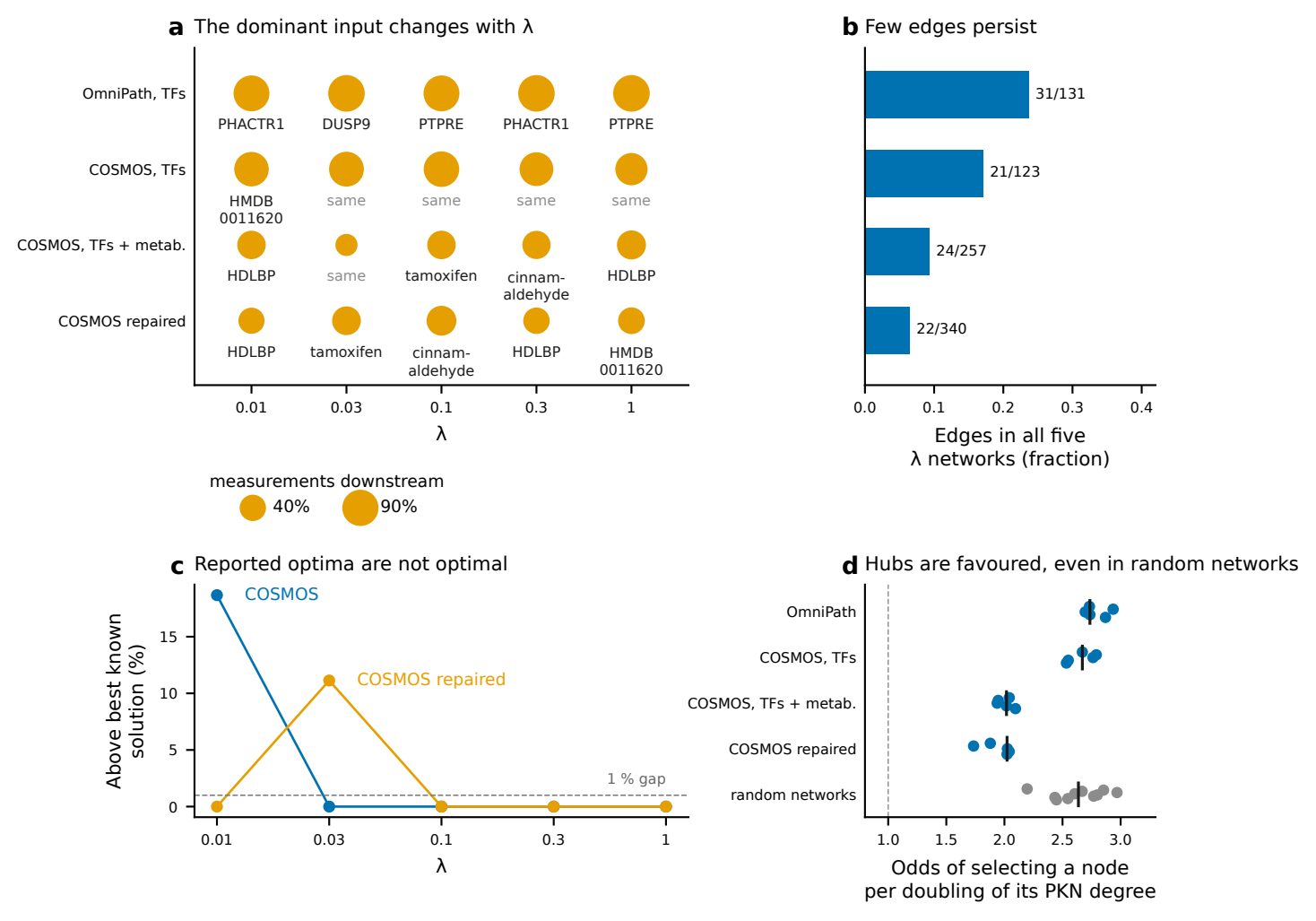
a The dominant input changes with λ
OmniPath, TFs
COSMOS, TFs
COSMOS, TFs + metab.
COSMOS repaired
PHACTR1
DUSP9
PTPRE
PHACTR1
PTPRE
HMDB 0011620
same
same
same
same
HDLBP
same
tamoxifen
cinnam-aldehyde
HDLBP
HDLBP
tamoxifen
cinnam-aldehyde
HDLBP
HMDB 0011620
0.01
0.03
0.1
0.3
1
λ
measurements downstream
40%
90%
b Few edges persist
31/131
21/123
24/257
22/340
0.0
0.1
0.2
0.3
0.4
Edges in all five λ networks (fraction)
c Reported optima are not optimal
COSMOS
COSMOS repaired
1 % gap
Above best known solution (%)
15
10
5
0
0.01
0.03
0.1
0.3
1
λ
d Hubs are favoured, even in random networks
OmniPath
COSMOS, TFs
COSMOS, TFs + metab.
COSMOS repaired
random networks
1.0
1.5
2.0
2.5
3.0
Odds of selecting a node per doubling of its PKN degree